

Spatiotemporal Analysis of Davinci RNA-seq

- Pathway Biology of Normal Porcine Wound Healing:
 - Characterize Porcine genetic pathway activation underlying normal wound healing.
 - Identify biological programs associated with healthy porcine wound healing.
 - Leverage **Non-Negative Matrix Factorization** (NMF) to discover latent pathway activity.
 - [GeneOntology.org](https://www.geneontology.org) used to annotate NMF meta-genes.
 - Methodology follows [Fukushima et al.](#)

Non-Negative Matrix Factorization Overview

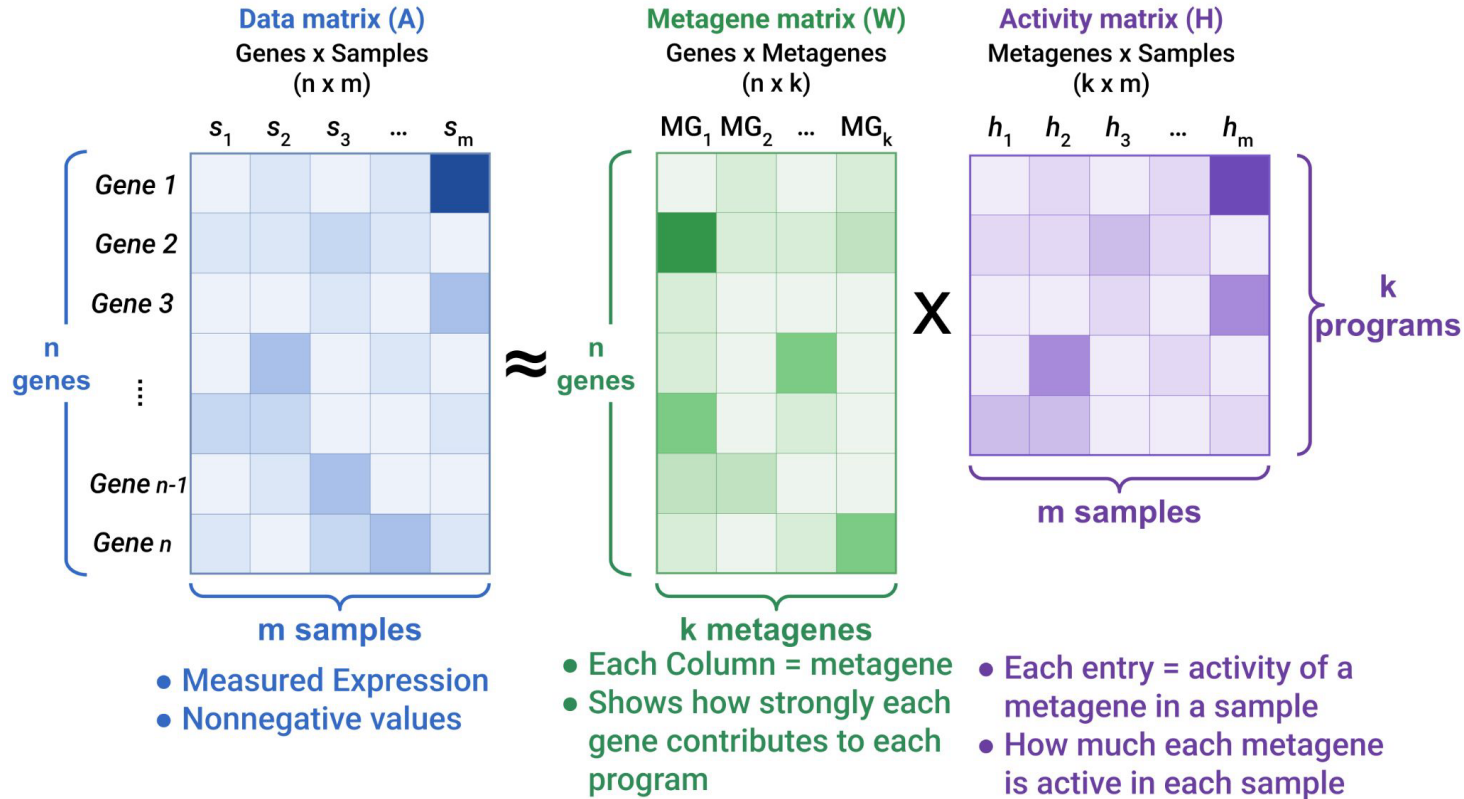


Figure 1: Overview of Non-Negative Matrix Factorization.

Non-Negative Matrix Factorization Results

- Healing-Specific Metagenes represent:
 - **Program 2:** Skin Morphogenesis & Development.
 - **Program 3:** Inflammatory Defense-Stress Response.
 - **Program 4:** Collagen, Cellular, Tissue Development.
 - **Program 6:** Cornification, Keratinization, Skin Barrier.

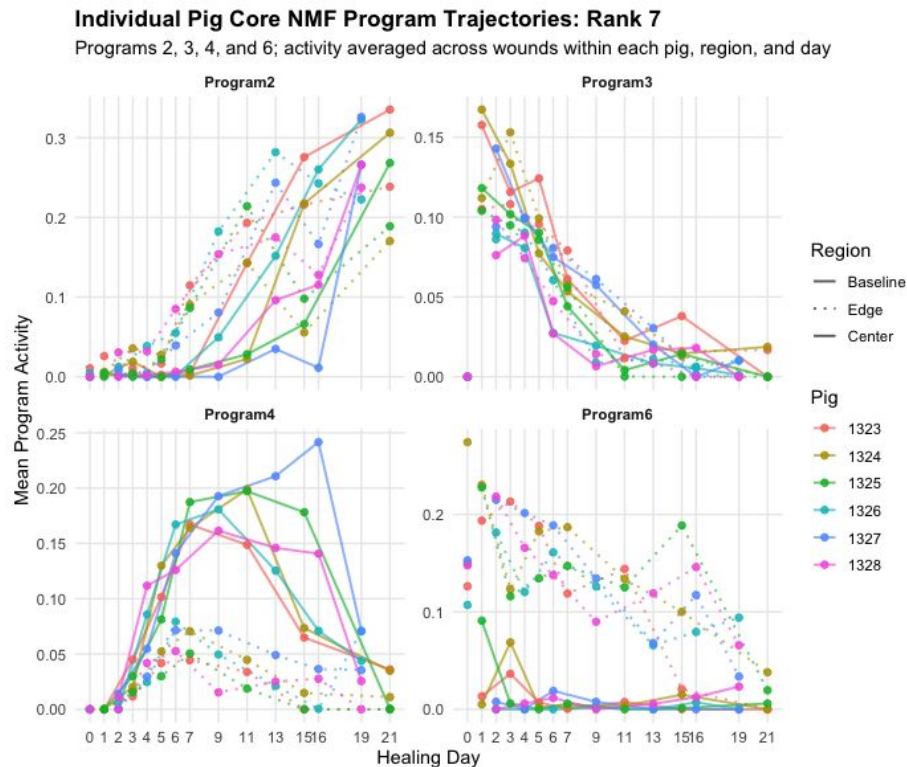
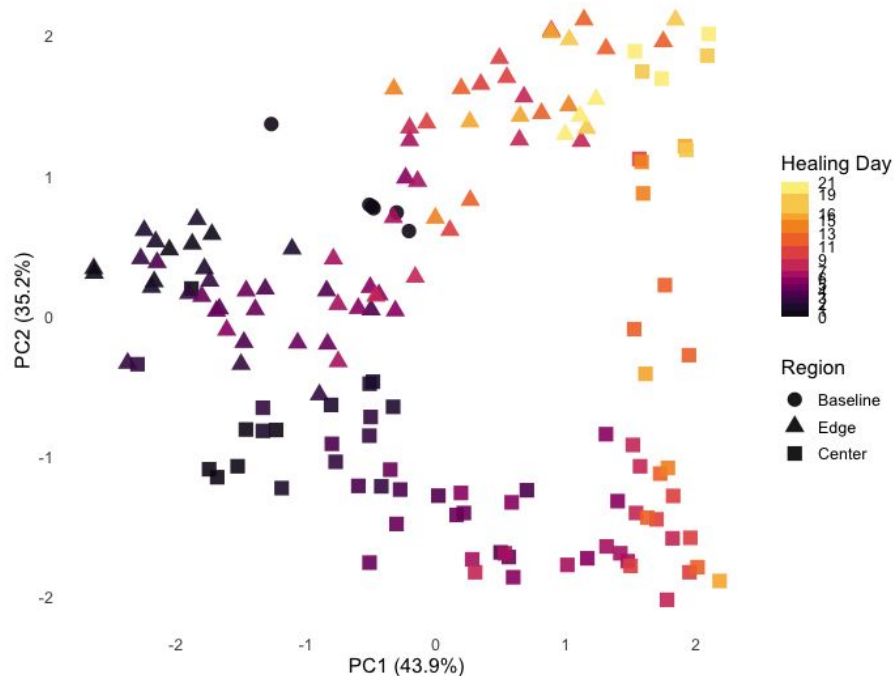


Figure 3: Temporal dynamics of 4 Healing-specific metagenes over 21 days.

PCA and UMAP of Healing-Specific Metagenes

PCA of Core Porcine Wound-Healing Programs

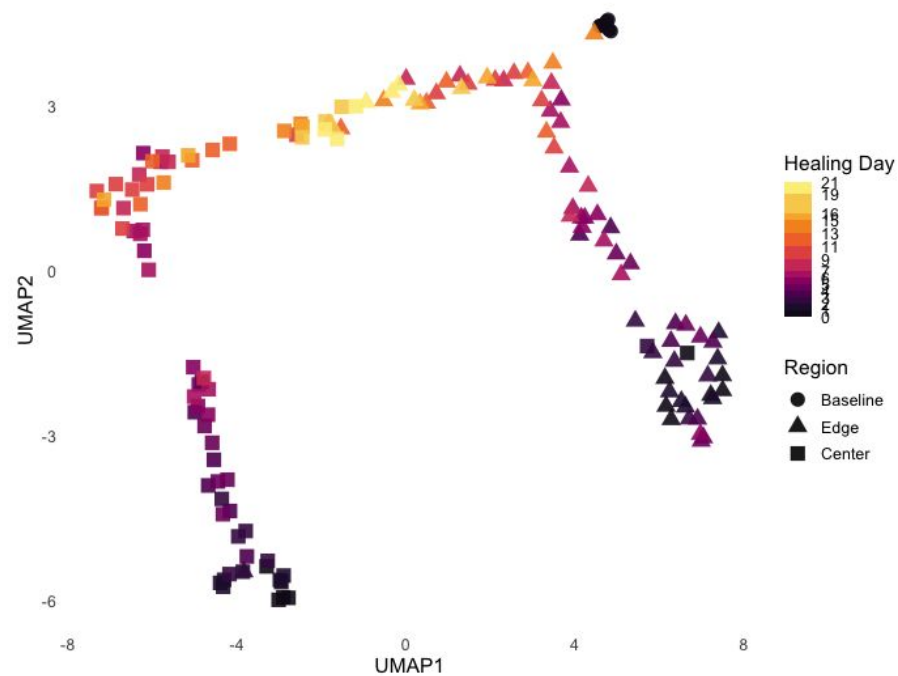
Rank-7 NMF Programs 2, 3, 4, and 6; composition-associated programs excluded



A

UMAP of Core Porcine Wound-Healing Programs

Rank-7 NMF Programs 2, 3, 4, and 6; composition-associated programs excluded



B

Figure 3: (A) PCA and (B) UMAP of four Healing-Specific metagenes over 21 days. NMF healing programs trace an organized trajectory through latent space, suggesting that wound repair can be represented by a compact set of molecular coordinates.